

Letterbox with Letter-Counting Facility



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Here is a circuit that starts counting when you insert a letter in the letterbox at your home or office. It is designed to save your time from going to the letterbox to check if there are letters inside. The number of letters present in the box is indicated by a seven-segment display. The block diagram of the circuit is shown in Fig. 1.

Circuit and working

Fig. 2 shows the circuit diagram of

the electronic letterbox with letter-counting facility. It is built around a white LED (LED1) and an LDR (LDR1), popular timer NE555 (IC1) in monostable mode, a counter, seven-segment driver CD4033 (IC2) and a few other components. LED1 and LDR1 together work as a sensor.

The resistance of LDR1 changes in accordance with the intensity of light incident on it. When light from LED1 falls on LDR1, its resistance is low.

Voltage at pin 2 of IC1 depends on the light falling on LDR1. In the dark, voltage at pin 2 is low, and vice-versa.

When a letter is inserted into the letterbox, it passes between LED1 and LDR1. This interrupts the light falling from LED1 to LDR1. As a result, resistance of LDR1 increases. This change in resistance provides

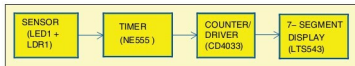


Fig. 1: Block diagram of the circuit

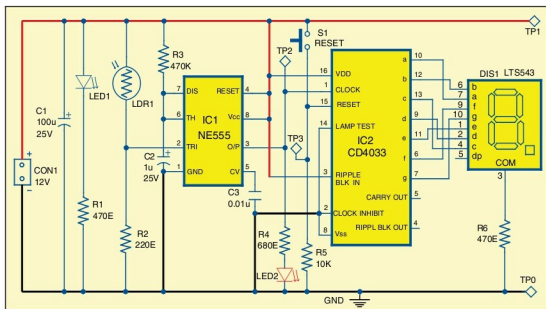


Fig. 2: Circuit diagram of the electronic letterbox with letter-counting facility

PARTS LIST

Semiconductors:

IC1	- NE555 timer
IC2	- CD4033 counter-cum-driver
LED1	- 5mm white LED
LED2	- 5mm LED
DIS1	- LT5543 seven-segment

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1, R6	- 470-ohm
R2	- 220-ohm
R3	- 470-kilo-ohm
R4	- 680-ohm
R5	- 10-kilo-ohm

Capacitors:

C1	- 100µF, 25V electrolytic
C2	- 1µF, 25V electrolytic
C3	- 0.01µF ceramic disk

Miscellaneous:

CON1	- 2-pin connector terminal
S1	- Tactile switch
LDR1	- LDR (light dependant resistor)
	- 12V regulated power supply

wave pulse at its output pin 3. This pulse acts as clock input for the counter and display driver CD4033 (IC2). Output pins of IC2 are connected to various segments a, b, c, d, e, f and g pins of seven-segment display (DIS1) as shown in the circuit. Its common pin 3 is connected

to ground through current-limiting resistor R6. Alternatively, you can also provide a resistor for each segment after removing resistor R6. Seven-segment display DIS1 displays the number of letters present in the box up to nine.

When a letter is delivered to the letterbox, LED2 momentarily glows, which indicates that a letter is received